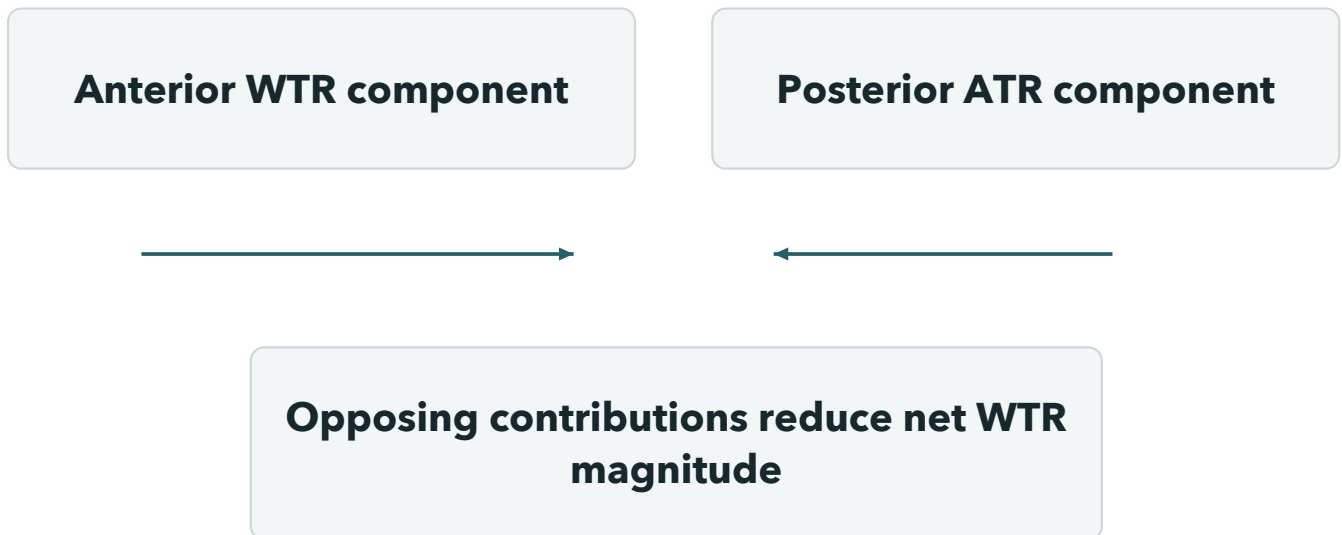


Astigmatic and binocular refractive planning

Part 2 | Anterior cylinder is not the whole cornea

Posterior astigmatism can change the total regular-cylinder target.



Original conceptual vectors: posterior ATR contribution can offset anterior WTR magnitude.

Use a validated total-corneal strategy and assess measurement quality. Ignoring posterior ATR can overestimate WTR correction when components oppose. Irregularity, incision-induced change and implant rotation also contribute to residual cylinder. This example does not supply a universal correction factor.

A toric IOL corrects predictable regular cylinder

Steepness alone does not establish suitability in ectasia.

Requirement	Why it matters
Stable central axis and magnitude	Improves prediction reliability
Regular central pattern	Matches the optic correction
Repeatable measurements	Reduces planning from artefact
Future contact-lens needs	May alter the preferred strategy

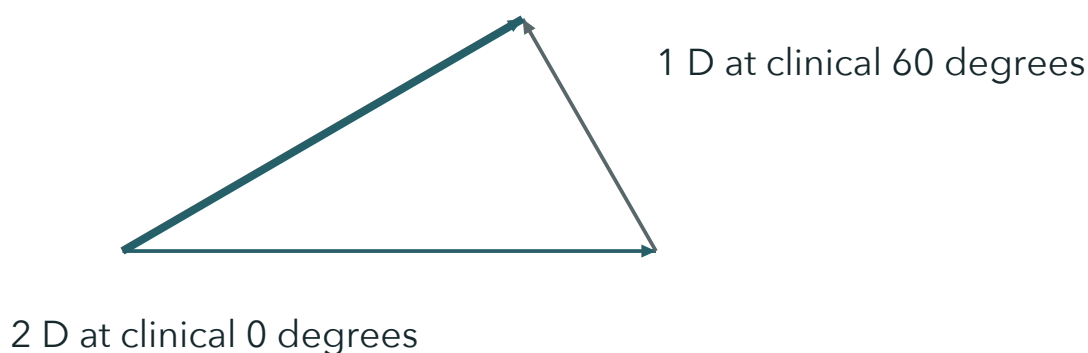
Assess alignment, total-corneal cylinder and the likelihood of later shape change. A toric implant cannot remove all higher-order ectatic aberrations. Lens rotation and the corneal target are different failure mechanisms; investigate both when residual astigmatism is greater than expected.

Cylinder axes require double-angle arithmetic

Scalar addition ignores the direction of astigmatic change.

Result: 1.73 D at clinical 15 degrees

Vector angles: 0, 120 and 30 degrees



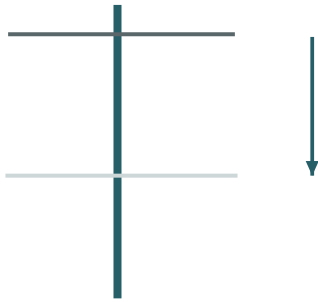
Original worked vector: clinical axes are doubled for summation and halved for the result.

For the stated model, 2 D at clinical 0 degrees plus 1 D at clinical 60 degrees gives components (1.50, 0.866), magnitude 1.73 D and clinical axis 15 degrees. Specify cylinder convention and the represented vectors before applying arithmetic; a corneal-power axis and a spectacle-cylinder axis are not interchangeable labels.

Unequal blur, image size and prism are different burdens

A patient can tolerate distance correction yet struggle in downgaze.

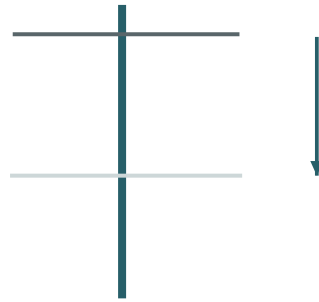
Right: -6 D



1 cm below centre

6 prism D base down

Left: -2 D



1 cm below centre

2 prism D base down

Relative difference: 4 prism D base down on right

Original worked example: 1 cm downgaze through -6 D and -2 D produces unequal base-down prism.

Compare symptoms at distance and near. Differential spectacle prism depends on the viewing point, while optical or retinal aniseikonia concerns perceived image size. In this minus-lens example, the relative imbalance is 4 prism dioptres base down on the right; conventional slab-off adds 4 base up to that lower lens portion.

Monovision needs a binocular trial

Ocular dominance alone does not predict tolerance.

- Assess alignment, fusion, stereo and visual demands.
↓
- Discuss unequal blur and range-versus-quality tradeoffs.
↓
- Use a reversible simulation when feasible.
↓
- Select targets from tolerance, not dominance alone.

Previous strabismus or limited fusion changes the risk discussion. A contact-lens simulation supplies practical information but cannot guarantee every later surgical symptom. Plan the fellow eye and future correction together; a technically successful monocular target can still create an unacceptable binocular result.